Micron NUS-ISE Business Analytics Case Competition 2026

Competition Preparation Material

Contents

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The Micron NUS-ISE Business Analytics Case Competition 2026 organizers appreciate the commitment of all participants to maintaining the integrity of the competition and ensuring fair play among teams. We wish all participants the best of luck in their preparations and look forward to an engaging and successful competition.

What is ISE?

The core principle of Industrial and Systems Engineering (ISE) centers around solving problems in multiple domains, including logistics, business, engineering, etc., backed by scientific approaches in data analytics, systems modelling, decision-making and management. Find out more: https://cde.nus.edu.sg/isem/what-is-ise/

DATA ANALYTICS AND DECISION MAKING

Deriving solutions that maximize productivity and minimize cost by applying data analytics and systems modelling, backed by a strong foundation in decision analysis and related economic concepts. Subjects include *Operations Research, Machine Learning, Decision Analysis, Mathematics, Probability and Statistics*.

COMPUTATIONAL AND MODELLING SKILLS

Applying programming and computational tools and techniques to analyze and solve complex problems. These include Python, R, MS Excel VBA, AutoMod/@Risk (simulation), STELLA (systems dynamics), AIMMS (optimization) and DPL (decision analysis). Subjects include *Simulation, Programming, Data Structures*.

SYSTEMS ENGINEERING AND ANALYSIS

A systematic approach to design, development, and innovation of real-world systems integrating people, technology, information, and resources, while accounting for global, societal, environmental, and economic contexts. Subjects include *Systems Thinking and Dynamics, Supply Chain Systems, Manufacturing Systems, Service Systems.*

Keen to know more? Find out more about **Industrial and Systems Engineering at the National University of Singapore** https://cde.nus.edu.sg/isem and learn more about partner **Micron** https://sg.micron.com/.

What is BACC?

The Micron NUS-ISE Business Analytics Case Competition (BACC) is an annual competition organized by the Industrial and Systems Engineering Club at the National University of Singapore since 2012.

Participants come together to solve real-world business problems set by our industry partners. After receiving the case question, they must race against time to understand the problem, consider various scenarios, and pain points, and finally make a recommendation backed by sound quantitative analysis to our panel of judges.

No prior experience is expected of participants. Although experience with data analysis, wrangling, visualization and even programming may be useful, such experience is not needed to participate.

Micron Partnership with NUS-ISE

The BACC Organizers for the 2026 case competition are pleased to announce Micron Technology as their partner for the event. Micron Technology is a global leader in semiconductor manufacturing, specializing in memory and storage solutions for a diverse range of applications, including computing, networking, and automotive industries.

This competition is organized by the NUS ISE Club and is supported by the NUS Department of Industrial Systems Engineering & Management. For more information or inquiries, contact nus.isecasecomp@gmail.com.

Competition Details

The competition features two categories: the Undergraduate Category for NUS students and the Pre-Uni Category for pre-university students in Singapore and NSFs. Teams of four or five members can sign up, and the registration deadline is 19th February 2026. Prizes for each category include SGD 1200 for the 1st place, SGD 1000 for the 2nd place, SGD 800 for the 3rd place, and four consolation prizes of up to SGD 400.

Additionally, there will be a Lucky Draw on the last day, open to all participants who make it to the finals. Lucky draw prizes are as follows: 1) iPad A16 11-inch, 2) Airpods 3 Pro, 3) Logitech Keyboard. The competition timeline comprises of the opening day and question release on 21st February 2026, the submission deadline on 22nd February 2026, the announcement of finalists on 26th February 2026, and the final day presentations by finalists on 28th February 2026. The venues are the online platform Zoom for opening day and question release, and the NUS campus for the final day presentations. More details will be provided closer to the date.

ISEM Department Recommended Reading List

1. **"Operations Research Applications Algorithms" by Wayne L. Winston (4th Edition):**
   This comprehensive text delves into the field of operations research, providing an in-depth exploration of applications and algorithms. Winston's approach is known for its practicality and real-world relevance, making it a valuable resource for understanding and applying operations research concepts.

2. **"Engineering Economy" by William G. Sullivan and Elin M. Wicks (16th Edition):**
   Sullivan and Wicks' "Engineering Economy" is a widely used textbook that focuses on the economic aspects of engineering projects. It covers essential topics such as cost estimation, financial analysis, and decision-making, equipping readers with the necessary skills to assess and optimize engineering investments.

3. **"Data Analytics: Become a Master in Data Analytics" by Richard Dorsey (2017):**
   Richard Dorsey's book serves as a comprehensive guide for mastering data analytics. It covers a wide range of topics, including data collection, processing, analysis, and interpretation. The book is suitable for both beginners and those looking to deepen their understanding of data analytics techniques.

Past Training Workshops

Online workshops conducted by the NUS ISEM Department:

1. **"Simulation Workshop" by Dr Tan Chin Hon:**
   Link: http://tinyurl.com/NUSISEMSimulationWorkshop
   The workshop explores the use of simulation to enhance operational decision-making. Simulation's wider applications in decision-making, including decisions related to hiring manpower or acquiring equipment, provide companies with the ability to pinpoint optimal actions without incurring significant costs.

2. "**Introduction to Linear Optimization" by Prof Napat Rujeerapaiboon:**
   Link: http://tinyurl.com/NUSISEMLinearOptimization
   Linear programming helps make optimal decisions by representing real-world problems with linear equations. It involves maximizing or minimizing an objective while considering limitations like resource availability. This powerful tool finds applications in many industries, from manufacturing to logistics.

3. "**Optimization using MS Excel Solver" by Dr Tan Chin Hon:**
   Link: http://tinyurl.com/NUSISEMOptimizationExcelSolver
   This workshop explains optimization as finding the best solution within constraints and outlines the steps: defining variables, specifying the objective function, setting constraints, and using Excel Solver for optimization.

4. "**Machine Learning Workshop" by Dr Tan Chin Hon:**
   Link: http://tinyurl.com/NUSISEMMachineLearningWorkshop
   The workshop delves into the fundamentals of machine learning and its practical applications in addressing real-world challenges. A hands-on example is provided, illustrating the process of training a machine learning model for image classification.

5. "**Introduction to Simulation & Modelling Workshop" by Yeo Keng Swee, Steve:**
   Link: http://tinyurl.com/NUSISEMSimulationModelling
   The workshop covers topics such as digital twin simulation, building simulation models in FlexSim, and utilizing simulation modelling for system optimization.

2025 Past Year Competition Question

Disclosure

For avoidance of doubt, all figures and numbers used in the questions provided are strictly arbitrary and have no reference whatsoever to Micron Technology, Inc.

Introduction

Micron Technology, a leading memory and storage solutions provider, has been strategically expanding its manufacturing capabilities to meet the growing demand. The company has developed a roadmap to address this future memory demand over the next decade by investing over $150 billion globally in manufacturing and R&D. This includes advancements in DRAM and NAND technologies, which are essential for various applications like AI, 5G Technology, and Data Centers.

<u>Demand Dynamics</u>

The demand for Micron's products is driven by several key factors:

- **Artificial Intelligence (AI)**: The rapid growth in AI applications, particularly in data centers, has significantly increased the demand for high-performance memory solutions.
- **5G Technology**: The expansion of 5G networks is boosting the need for advanced memory and storage solutions in smartphones and other connected devices.
- **Data Centers**: With the rise of cloud computing and big data, data centers require more robust and efficient memory solutions.

Background

Market Trends

Micron effectively manages the supply-demand balance by leveraging AI server demand and supply reductions to drive pricing improvements. The company has also seen increased interest from customers in securing long-term agreements due to expectations of tight supply and rising prices.

Problem Statement

Despite these positive trends, Micron faces several significant challenges which can be summarized into the following:

1. **Cyclical Demand**: The semiconductor industry is known for its cyclical demand, where periods of high demand can be followed by oversupply. This volatility affects Micron's revenue and profitability.

2. **Geopolitical Tensions**: Global political issues, such as trade restrictions and export controls, can impact Micron's operations and market access. These tensions can disrupt supply chains and affect the availability of critical materials.

3. **Intense Competition**: Micron competes with major players like Samsung, SK Hynix, and Western Digital. This competition requires continuous investment in research and development to stay ahead in terms of technology and innovation.

4. **Technological Advancements**: Rapid advancements in technology necessitate significant capital investment. Micron must continuously innovate to keep pace with emerging technologies like AI, IoT, and 5G, which drive demand for advanced memory solutions.

5. **Supply Chain Issues**: The semiconductor supply chain is complex and can be affected by various factors, including natural disasters, pandemics, and logistical challenges. Ensuring a stable supply chain is crucial for Micron's operations.

In this Business Analytics Case Competition, we will take a closer look into the elements of capacity planning based on the Total Addressable Market (TAM) to sustainably navigate the challenging environment especially the industry's cyclical nature, while maintaining the company's competitive edge and ensuring its profits are maximized.

Challenge

Question 1:

Part a)

Micron's executive team has determined an optimal business strategy to be executed from 2026 to 2027. During this period, the Total Addressable Market (TAM) for Micron for memory bits has been forecasted with a range and a fixed contribution margin per GB as detailed in Table 1. Micron produces different generations of products together at the Singapore Fab, which are currently labelled Node 1, Node 2 and Node 3. Each product has its GB per wafer and Yield information outlined in Table 2.

Table 1: Summary of TAM and Selling Price per GB from 2026 to 2027 (breakdown per quarter)

Quarter	Q1'26	Q2'26	Q3'26	Q4'26	Q1'27	Q2'27	Q3'27	Q4'27
TAM (±2 billion GBs)	21.8	27.4	34.9	39.0	44.7	51.5	52.5	53.5
Contribution margin per GB	\$0.002	\$0.002	\$0.002	\$0.002	\$0.002	\$0.002	\$0.002	\$0.002

Table 2: Summary of each product's GB per wafer and Yield

Quarter		Q1'26	Q2'26	Q3'26	Q4'26	Q1'27	Q2'27	Q3'27	Q4'27
Node 1	GB per wafer	100k	100k	100k	100k	100k	100k	100k	100k
	Yield	98%	98%	98%	98%	98%	98%	98%	98%
Node 2	GB per wafer	150k	150k	150k	150k	150k	150k	150k	150k
	Yield	60%	82%	95%	98%	98%	98%	98%	98%
Node 3	GB per wafer	270k	270k	270k	270k	270k	270k	270k	270k
	Yield	20%	25%	35%	50%	65%	85%	95%	98%

As an industrial engineer, you have been tasked by Micron's executive team to come up with a feasible product mix for the different generations that fulfills the business plan as summarized in Table 1. This consists of designing a loading (Table 3) that details the number of wafers that needs to be started each week for each product generation for each quarter.

Table 3: Loading profile of different product generations (Nodes 1, 2 and 3)

Quarter	Q1'26	Q2'26	Q3'26	Q4'26	Q1'27	Q2'27	Q3'27	Q4'27
Node 1	12000							
Node 2	5000							
Node 3	1000							

These are the constraints and assumptions to be satisfied when designing the loading:

1. Your designed loading needs to provide sufficient GB output to fulfill the TAM within the forecasted range. No additional GB should be produced per quarter which will lead to excess inventory and adversely impact the market price of each GB.
2. You are given Q1’26 loading as the initial condition.
3. Each node’s loading is independent and cannot deviate more than 2.5k wafers from each quarter to the next. For example, node 2 has 5k volume in Q1’26, so its Q2’26 volume cannot be beyond the range of 5±2.5k.
4. Loading volume cannot be negative.

Below are also the standard definitions and calculation methodologies to determine whether the business plan is met:

Attribute	**Definition**
TAM per quarter	The total number of GB output that meets the quality standard to be shipped to the customers to meet demand. TAM is provided as total GB needed for each quarter.
Quarter	Each quarter consist of **13** weeks, with each year having 4 quarters
Loading	Loading is planned on a weekly basis, it is the number of wafers to be started each week at the Fab. The loading will be constant throughout all weeks in a quarter. i.e. Loading of **5k** for node 2 in Q1’26 means we will start 5k node 2 wafers for each of the 13 weeks in Q1’26, in total 65k node 2 wafers will be produced in Q1’26.
GB per wafer	This is the designed number of GB each perfectly fabricated wafer contains (GBpW)
Yield	No wafer can be perfectly fabricated. This is the percentage of GB that is expected to be successfully fabricated out of the total designed GB amount. As fabrication process matures, yield will increase
GB output per quarter	The final GB amount that is successfully fabricated from the designed loading. It is calculated as follows for each quarter: $GB\ output\ =\ 13\cdot \sum_{node} Loading\cdot GBpW\cdot Yield$

Part b)

Once the loading is determined, you will need to design a tooling plan for the workstations to produce the required number of wafers. The capacity governing equation is shown in Equation 1 which provides the tool requirement for each product based on a given loading. This tool requirement can be translated to Capital Expenditure (CAPEX) required to support the designed loading. The CAPEX information for 10 different workstations and their Utilization/Minute Load/initial available tool count is summarized in Table 4.

Equation 1:

$$Tool\ Requirement\ = \sum_{node} \frac{Loading \cdot Minute\ Load}{7 \cdot 24 \cdot 60 \cdot Utilization}$$

Table 4: Summary of 10 different workstations and their metrics

Workstation	A	B	C	D	E	F	G	H	I	J
Initial tool count for Q1'26	10	18	5	11	15	2	23	3	4	1
Utilization	78%	76%	80%	80%	76%	80%	70%	85%	75%	60%
CAPEX per tool	\$3.0M	\$6.0M	\$2.2M	\$3.0M	\$3.5M	\$6.0M	\$2.1M	\$1.8M	\$3.0M	\$8.0M
Node 1 Minute Load	4.0	6.0	2.0	5.0	5.0		12.0	2.1		
Node 2 Minute Load	4.0	9.0	2.0	5.0	10.0	1.8			6.0	
Node 3 Minute Load	4.0	15.0	5.4			5.8	16.0			2.1

(i) Use the loading profile in (a) to determine the tool requirement for each quarter.

(ii) Use the tool requirement calculated in (bi) to provide a quarterly breakdown of the total CAPEX incurred from additional tool purchase.

(iii) Calculate the net profit based on the selling price per GB in Table 1.

(iv) With reference from (bi) to (biii), determine if this is the best loading profile to maximize profits. You can consider adjusting the loading profile in (a) and tooling plan in (b).

Assumptions to be considered:

1. For each node's wafer to be produced, they need to run through all workstations that has non-zero minute load for that node. For example, node 3 need to run through workstations A, B, C, F, G, J and thus contribute to their tool requirement.
2. You can assume that tool requirement for each workstation is independent and only governed by Equation 1 and nothing else. All other considerations such as transport time

between workstations, queueing time, tool idle time, process sequence etc. are negligible or already captured under a variable in Equation 1.

   a. For example, workstation G's requirement in Q1'26 is: $\frac{12000*12+1000*16}{10080*0.7} = 22.66$, this is entirely independent of workstation H's requirement: $\frac{12000*2.1}{10080*0.85} = 2.94$

3. Enough tools must be purchased to fully cover the tool requirement for each workstation. For example, if tool requirement is 3.08, 4 tools must be available for production. For workstations with multiple nodes, requirement for each node should be summed before the tool requirement is determined.
4. The initial available tool count at Q1'26 are fully paid for and will not incur any CAPEX.

Attribute	**Definition**
Tool Requirement	The numbers of tools required to produce the specified loading. While exact tool requirement can be a decimal, the final required tool count is an integer rounded up from the exact decimal.
RPT	Recipe Processing Time. This is the time in minutes that is required to complete the recipe that is uniquely associated to a step
Minute Load	The number of minutes a wafer needs to be processed at each workstation. This is the sum of all RPT of the steps running in the workstation.
Utilization	The percentage of time that a tool in a workstation can be used for production
CAPEX requirement	Calculated by multiplying required tool purchases with each workstation's CAPEX per tool
Contribution margin	This is the selling price per GB minus the operating expense per GB, it represents the profit per GB sold before accounting for fixed CAPEX cost
Profit	$Profit = GB\ output \cdot Contribution\ margin\ per\ GB - Capex\ requiremen$

Part c)

In no more than 150 words, provide a justification on the choice of this loading.

Question 2:

In the previous question, the capacity governing equation assumes that the RPT is static. However, RPT varies from wafer-to-wafer in reality. This variability introduces some uncertainty into our capacity planning. Currently this static RPT number is assumed to be the **median** RPT of each step. Micron's executive team has tasked you to determine the risk associated with the plan when considering RPT variability and whether median is the best method to obtain an RPT value.

Part a)

To quantify the risk to your plan in Question 1, you will need to calculate the probability of not being able to fulfill the designed wafer output.

Scope of this study has been limited to workstations H, I, and J, running 1 step each for Nodes 1, 2, 3 respectively. A sample of their RPT has been collected and provided in the attached excel file.

Table 5: Workstation H, I, J's median RPT

Workstation	H	I	J
Initial tool count for Q1'26	3	4	1
Utilization	85%	75%	60%
Node 1 Median RPT	2.1		
Node 2 Median RPT		6.0	
Node 3 Median RPT			2.1

Assumptions to be considered:

1. The raw RPT data provided is a representative sample of the actual RPT distribution.
2. The actual RPT can be treated as a random distribution.
3. All wafers' RPT are independent of each other.
4. Not being able to meet the designed loading means the tool requirement of a particular week exceeds the tool available.
5. Designed wafer output is not met if any week in the planning horizon from Q1'26 to Q4'27 cannot meet the designed loading & loading of each week in a quarter cannot be varied.
6. Assume each week is 1 time block, you do not need to consider any other time blocks.
7. If no viable loading plan has been found in question 1, you can consider the probability of not meeting Q1'26's baseline loading plan only.

Table 6: Q1'26 loading plan & GB output

Quarter	Q1'26
Node 1	**12000**
Node 2	**5000**
Node 3	**1000**

Here are some guidelines to calculate the probability of not meeting your designed loading:

(i) As the RPT samples are representative, are you able to find the RPT's population mean and standard deviation?

(ii) What is the expected value for each RPT based on the population characteristic determined in (i)?

(iii) Governing equation of calculating tool requirement can be interpreted as the total time required to run the designed loading divided by the available tool time on a weekly basis.

$$Tool\ Requirement\ = \sum_{node} \frac{Loading \cdot Minute\ Load}{7 \cdot 24 \cdot 60 \cdot Utilization}$$

What is the expected tool requirement based on this?

(iv) However, expected value is still a static measure. How can we find the total time to process X number of wafers where each wafer's processing time is a random variable following the population characteristics determined in (ii)?

a. Are you able to create a simulation for this or some analytical method?

b. Can different types of populations use the same methodology?

(v) From assumption 4, not being able to meet design output of a week means our total processing time determined in (iv) is higher than total available tool time. What is the probability associated with this inequality?

(vi) As all weeks in a quarter have the same designed loading and tool available, are we able to use the answer in (v) to find the probability of at least 1 week in a quarter not meeting the designed loading?

(vii) After finding the answer to (vi) for all quarters, can we find the probability of at least 1 quarter not meeting the designed loading? This will give you the probability of not meeting your designed loading throughout the planning horizon.

(Word Limit: 400)

Part b)

Given your results in 2(a), discuss whether median is the best statistical measure for estimating RPT. If not, what do you propose is the best method to estimate RPT? Evaluate your proposed method against the current method of using the median.

(Word Limit: 300)

Part c)

How would your answer in Q1 change, given the new insights gained from 2(a) and 2(b)? Discuss the risks associated with RPT variability. You may use the pointers stated below as reference:

(i) Will the projected profits be impacted?
(ii) What risk level is acceptable?
(iii) How should you change the planned loading or tool purchase to reduce the risk?

(Word Limit: 300)